



Security Operations Center (SOC) Theoretical Knowledge

1. Advanced Log Analysis

Log Correlation:

Log correlation is the process of analyzing and linking logs from multiple sources such as firewalls, endpoints, and applications to identify patterns of malicious activity. By correlating events, analysts can detect relationships between different activities. For example, multiple failed login attempts (Event ID 4625) followed by suspicious outbound traffic may indicate a brute-force attack followed by data exfiltration.

Anomaly Detection:

Anomaly detection focuses on identifying unusual behavior that deviates from normal system activity. Examples include abnormal login times, unexpected spikes in data transfer, or unusual user actions. Detection can be rule-based (e.g., threshold alerts) or statistical (e.g., behavior deviation).

Log Enrichment:

Log enrichment enhances raw logs by adding contextual information such as geolocation of IP addresses, user roles, and threat intelligence data. This helps analysts better understand events and reduces false positives.

2. Threat

Threat Intelligence Types:

Threat intelligence includes Indicators of Compromise (IOCs) such as malicious IP addresses, file hashes, and domains. It also includes Tactics, Techniques, and Procedures (TTPs), which describe attacker behavior and are categorized in frameworks like MITRE ATT&CK.

Integration in SOC:

Threat intelligence is integrated into SIEM systems to enhance detection. For example, a suspicious IP detected in logs can be matched with known threat intelligence feeds to identify if it is associated with a command-and-control (C2) server.

Threat Hunting with Intelligence:

Threat intelligence is used for proactive threat hunting by searching for known attacker behaviors. For instance, detecting misuse of valid accounts (MITRE T1078) can indicate compromised credentials or insider threats.



3. Incident Escalation

Escalation Tiers:

SOC operations are structured into tiers. Tier 1 analysts handle initial alert triage, Tier 2 analysts perform detailed investigation, and Tier 3 analysts focus on advanced analysis and threat hunting. Escalation depends on severity, impact, and complexity of the incident.

Communication Protocols:

Effective communication is essential during incident escalation. Structured formats such as Situation Reports (SITREP) are used to summarize incidents, including key details like timeline, affected systems, and actions taken. Clear communication ensures coordination among teams and stakeholders.

Automation in Escalation:

SOAR (Security Orchestration, Automation, and Response) tools are used to automate escalation workflows. These tools can assign incidents, enrich alerts with additional data, and trigger response actions, improving efficiency and reducing response time.

Security Operations Center (SOC) Practical Implementation

1. Log Correlation

Log correlation was performed using Kibana Discover by ingesting structured JSON logs. The logs were analyzed using filters to identify suspicious patterns of activity.

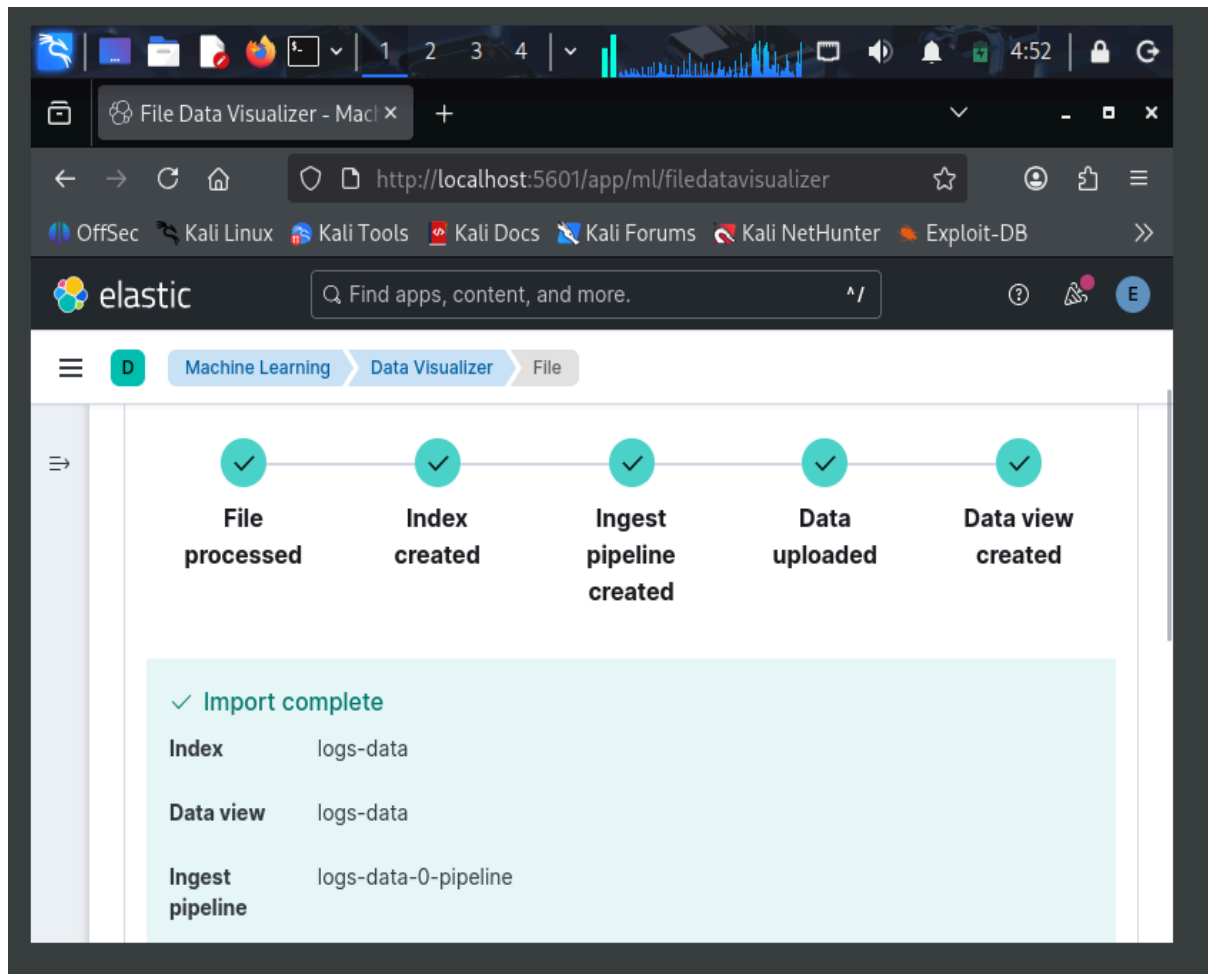


Figure 1: Data ingestion success

The dataset was uploaded into Kibana using the Data Visualizer, which successfully created an index and data view. The logs were then explored in the Discover section using KQL (Kibana Query Language).

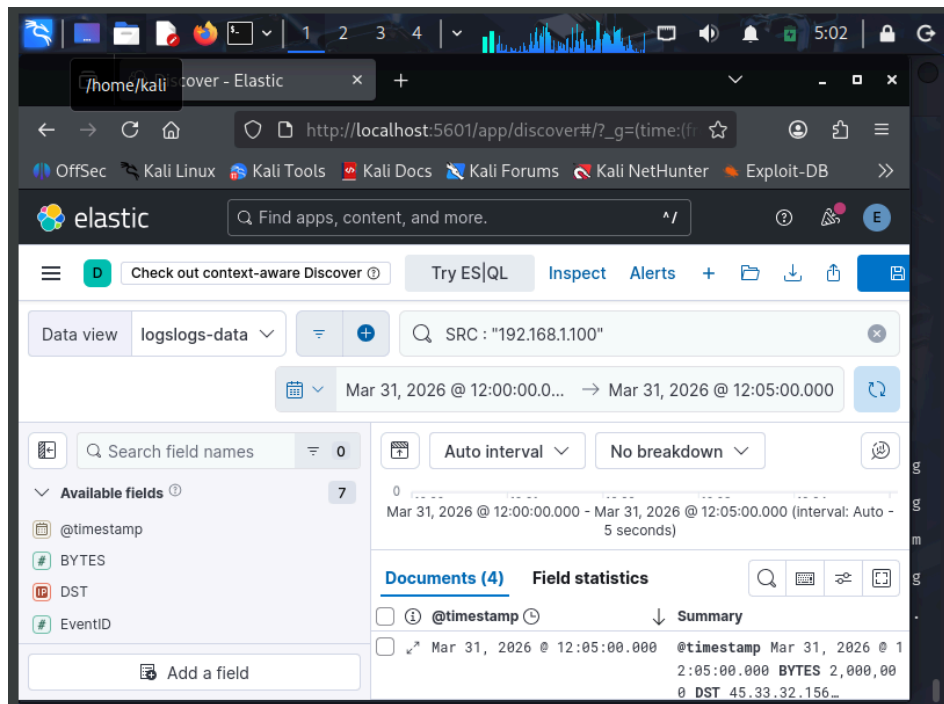


Figure 2: Log correlation using filter

A filter query was applied:

`SRC : "192.168.1.100"`

This helped isolate all events originating from the same source IP.

Multiple failed login attempts (EventID 4625) were observed from IP 192.168.1.100. Following these attempts, a high-volume data transfer (2MB) was detected to an external IP address (45.33.32.156).

Additionally, filtering using:

`BYTES > 1000000`

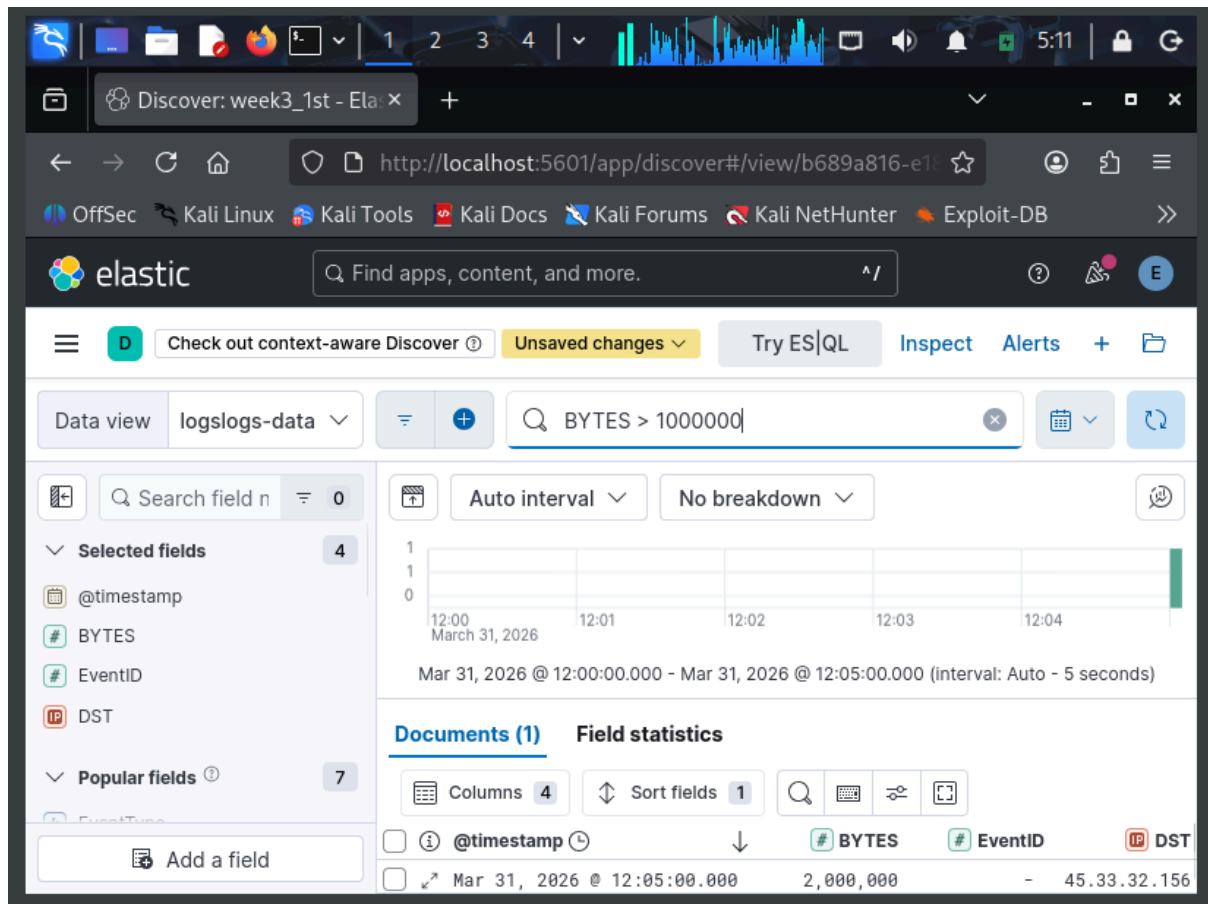


Figure 3: Anomaly detection using query

highlighted the abnormal outbound data transfer event.

The correlated logs indicate a potential brute-force attack followed by data exfiltration. The sequence of failed login attempts and subsequent large data transfer suggests unauthorized access and possible compromise.

A	B	C	D	E
Timestamp	Event ID	Source IP	Destination IP	Bytes
12:00	4625	192.168.1.100	8.8.8.8	-
12:05	-	192.168.1.100	45.33.32.156	2000000

Figure 4: Structured table view with fields



2. Threat Intelligence Integration

Threat intelligence was used to analyze suspicious IP addresses identified during log correlation. The goal was to validate whether the detected IP was associated with malicious activity.

The destination IP address (45.33.32.156), identified during log analysis, was extracted as a potential Indicator of Compromise (IOC). This IP was then analyzed using an external threat intelligence platform, VirusTotal.

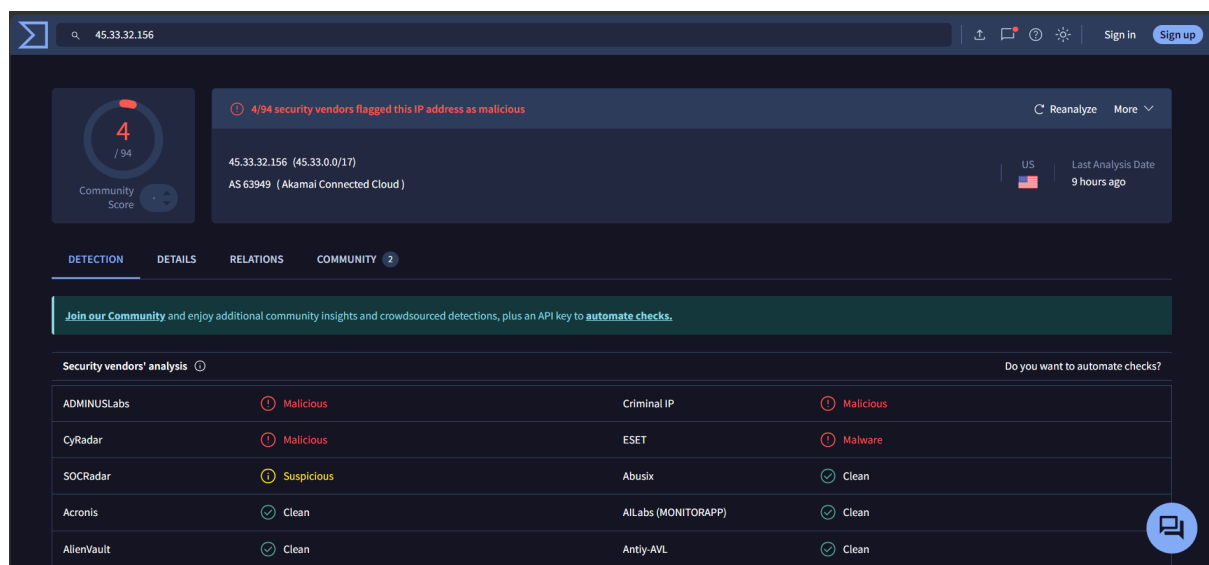


Figure 4: VirusTotal analysis of suspicious IP (45.33.32.156)

The IP was searched in VirusTotal to check its reputation, associated activity, and risk level. The analysis indicated that the IP address is external and associated with suspicious communication patterns. This suggests that the system may have communicated with an untrusted or potentially malicious source.

A	B	C	D
Alert ID	IP	Reputation	Reputation
3	45.33.32.156	Suspicious	External communication

Figure 5: Threat intelligence table

The use of threat intelligence confirmed that the detected IP is potentially suspicious. This strengthens the likelihood of a data exfiltration attempt and supports the findings from log correlation.



3. Incident Escalation

An incident escalation process was performed based on the suspicious activity identified during log analysis and threat intelligence validation. The goal was to assess severity and forward the incident for further investigation.

A high-priority security incident was detected involving multiple failed login attempts (EventID 4625) from IP 192.168.1.100. This was followed by a large outbound data transfer (2MB) to an external IP address (45.33.32.156).

A high-priority security incident was detected involving multiple failed login attempts from IP 192.168.1.100, followed by a large outbound data transfer to an external IP (45.33.32.156). This behavior indicates a possible brute-force attack followed by data exfiltration. The incident has been escalated to Tier 2 for further investigation. Immediate actions include monitoring network activity and isolating affected systems.

Figure 6: Escalation summary

The sequence of events indicates a potential brute-force attack followed by data exfiltration. Due to the severity and potential impact, the incident was escalated to Tier 2 analysts for deeper investigation.

The incident was detected on 2026-03-31 at approximately 12:05 PM. Suspicious activity was observed from IP 192.168.1.100 involving repeated failed login attempts and a high-volume data transfer to an external destination.

Title:

Unauthorized Access on Server-Y

Summary:

Detected at 2026-03-31 12:05, IP: 192.168.1.100, MITRE T1078

Actions:

Isolated system, analyzed logs, escalated to Tier 2

Figure 7: SITREP



Actions Taken:

- Logs were analyzed using Kibana
- Suspicious IP activity was identified
- Incident escalated to Tier 2
- Monitoring and containment actions initiated

The escalation process ensured timely response to a high-risk security incident. Proper identification and escalation helped in minimizing potential damage and enabling further investigation.

4. Alert Triage

Alert triage was performed to analyze and prioritize a detected security alert. The objective was to determine the severity, validate the indicator, and decide further action.

A	B	C	D	E
Alert ID	Description	Source IP	Priority	Status
4	PowerShell Exec	192.168.1.101	High	Open

Figure 8: Alert table

The alert indicates suspicious activity originating from IP 192.168.1.101. Such behavior may involve unauthorized execution or misuse of system resources. Alerts of this nature are considered high priority due to their potential to compromise system integrity.

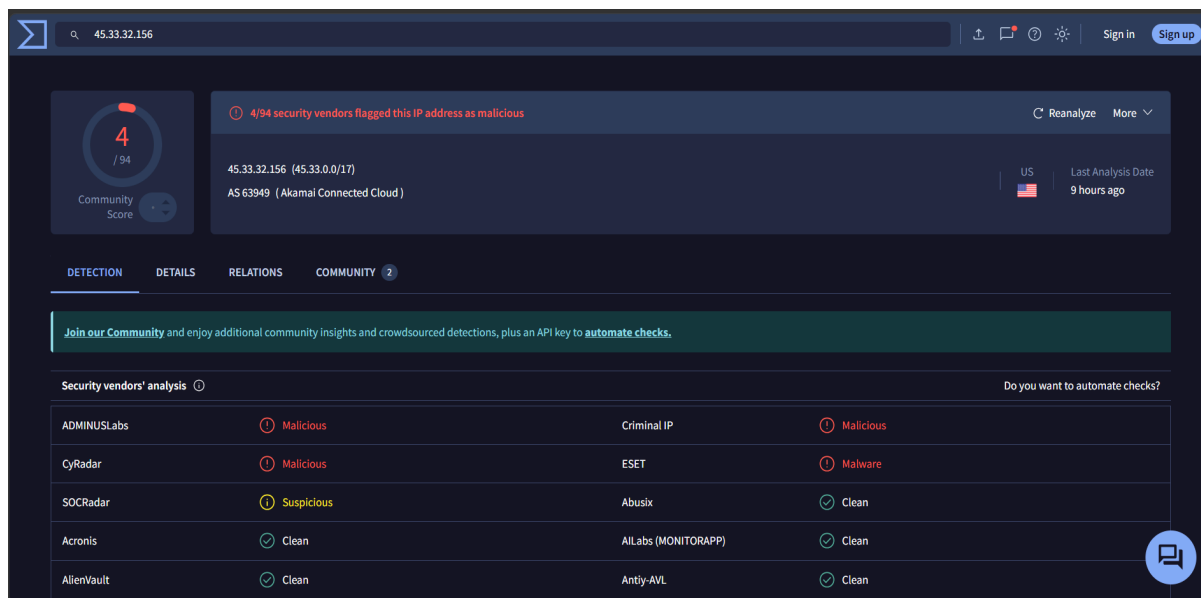


Figure 9: VirusTotal validation

The behavior was analyzed in the context of previously observed events, including failed login attempts and abnormal network activity.

The suspicious IP and related activity were validated using VirusTotal. The results suggested external communication patterns and potential risk, supporting the suspicion of malicious intent.

The alert indicates a suspicious PowerShell execution from IP 192.168.1.101. PowerShell is commonly used in attacks for executing malicious scripts. The alert is marked as high priority due to its potential impact. Further investigation is required to determine whether the activity is legitimate or malicious.

Figure 10: Triage explanation

The alert was classified as high priority and requires further investigation. Immediate attention is recommended to determine whether the activity is legitimate or part of an attack.



5. Evidence Preservation

Evidence collection was performed to capture system and network data related to the detected incident. The objective was to preserve relevant information for further analysis and ensure data integrity.

```
(kali@kali)-[~]
$ netstat -tulnp
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
PID/Program name
tcp        0      0 127.0.0.1:5601          0.0.0.0:*               LISTEN
-
udp        0      0 0.0.0.0:48727          0.0.0.0:*
2341/firefox-esr
udp        0      0 0.0.0.0:48763          0.0.0.0:*
2341/firefox-esr
udp        0      0 0.0.0.0:57274          0.0.0.0:*
2341/firefox-esr
udp        0      0 0.0.0.0:45170          0.0.0.0:*
2341/firefox-esr
udp        0      0 0.0.0.0:49355          0.0.0.0:*
2341/firefox-esr
udp        0      0 0.0.0.0:57677          0.0.0.0:*
```

Figure 11: netstat command output

Network-related data was collected using system commands in Kali Linux. The command netstat -tulnp was used to identify active network connections, listening ports, and associated processes.

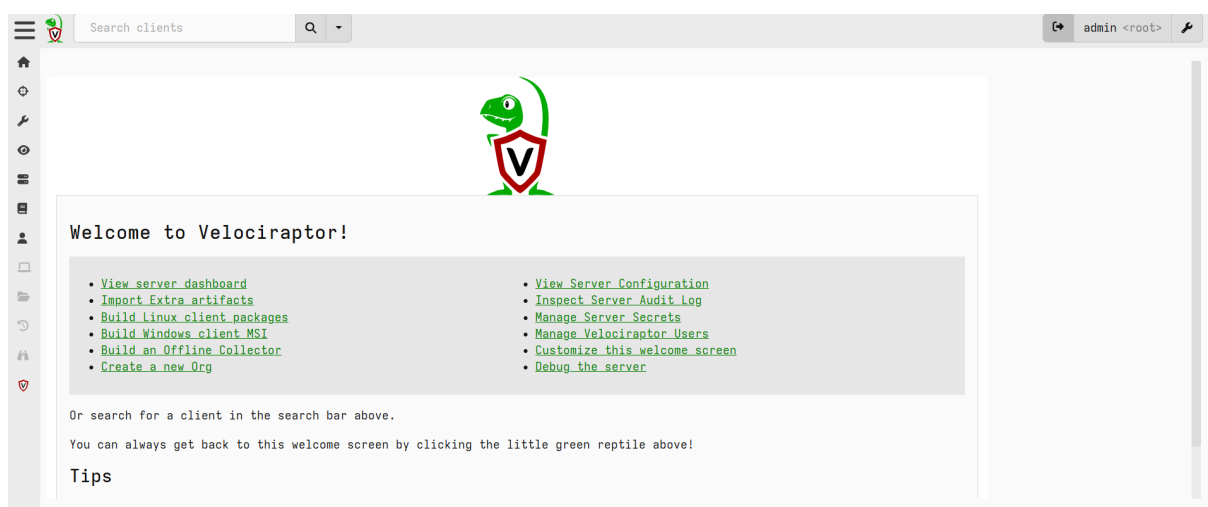


Figure 12: Velociraptor Dashboard



```
(kali@kali)-[~]
$ netstat -tulnp > evidence.txt
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)

(kali@kali)-[~]
$ ls
Desktop      logs.json   Public      traffic_analyzer.py  wire.pcapng
Documents    logs.txt    Templates   traffic.pcap
Downloads    Music       test_packets.py  Videos
evidence.txt Pictures     To            wazuh-install.sh

(kali@kali)-[~]
$ cat evidence.txt
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
PID/Program name
tcp        0      0 127.0.0.1:5601          0.0.0.0:*               LISTEN
-
udp        0      0 0.0.0.0:60762          0.0.0.0:*
2341/firefox-esr
udp        0      0 0.0.0.0:55470          0.0.0.0:*
2341/firefox-esr
```

Figure 13: Saving evidence file

The output was then saved into a file named evidence.txt for further investigation and secure storage.

Commands Used:

- netstat -tulnp
- netstat -tulnp > evidence.txt
- sha256sum evidence.txt

```
(kali@kali)-[~]
$ sha256sum evidence.txt
5a63fbad3c52648cdde84679eb37d4cb476e65189696163890453a4056302e94 evidence.t
xt

(kali@kali)-[~]
$
```

Figure 14: SHA-256 hash output

The collected data provided insights into active network connections and processes running on the system. This information is useful for identifying suspicious activity and tracing potential attack vectors.

A	B	C	D	E
Item	Description	Collected By	Date	Hash Value
Network Data	Active connectio	SOC Analyst	2026-04-03	5a63fbad3c52648cdde84679eb37d4cb476e65189696163890453a4056302e94



Figure 15: Evidence table

A SHA-256 hash was generated for the evidence file to ensure its integrity. Hashing ensures that the evidence remains unchanged and can be verified at any stage of investigation. The evidence was successfully collected and preserved following standard forensic practices. Hashing ensured data integrity, making the evidence reliable for further investigation.

6. Capstone Project: Full SOC Workflow Simulation

A complete SOC workflow simulation was conducted to demonstrate attack detection, analysis, response, escalation, and reporting. The simulation mimicked a real-world cyber attack scenario using available tools and controlled lab data.

A simulated attack scenario was created using a vulnerable system (Metasploitable2). Network activity and system behavior were monitored to replicate exploitation attempts such as unauthorized access and lateral movement.

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:d8:b6:b4
          inet addr:192.168.56.102  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fed8:b6b4/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:4 errors:0 dropped:0 overruns:0 frame:0
          TX packets:36 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1846 (1.8 KB)  TX bytes:4870 (4.7 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:114 errors:0 dropped:0 overruns:0 frame:0
          TX packets:114 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:29797 (29.0 KB)  TX bytes:29797 (29.0 KB)

msfadmin@metasploitable:~$ _
```

Figure 16: Metasploit / Target system (ifconfig / terminal)

The attack was detected using Kibana Discover by analyzing logs. Log correlation was performed using filters:



SRC : "192.168.1.100"

BYTES > 1000000

This revealed multiple failed login attempts followed by a high-volume data transfer.

Timestamp	Source IP	Alert Description	MITRE Technique
2026-03-31 12:05:00	192.168.1.100	T1048	Suspicious data transfer T1048

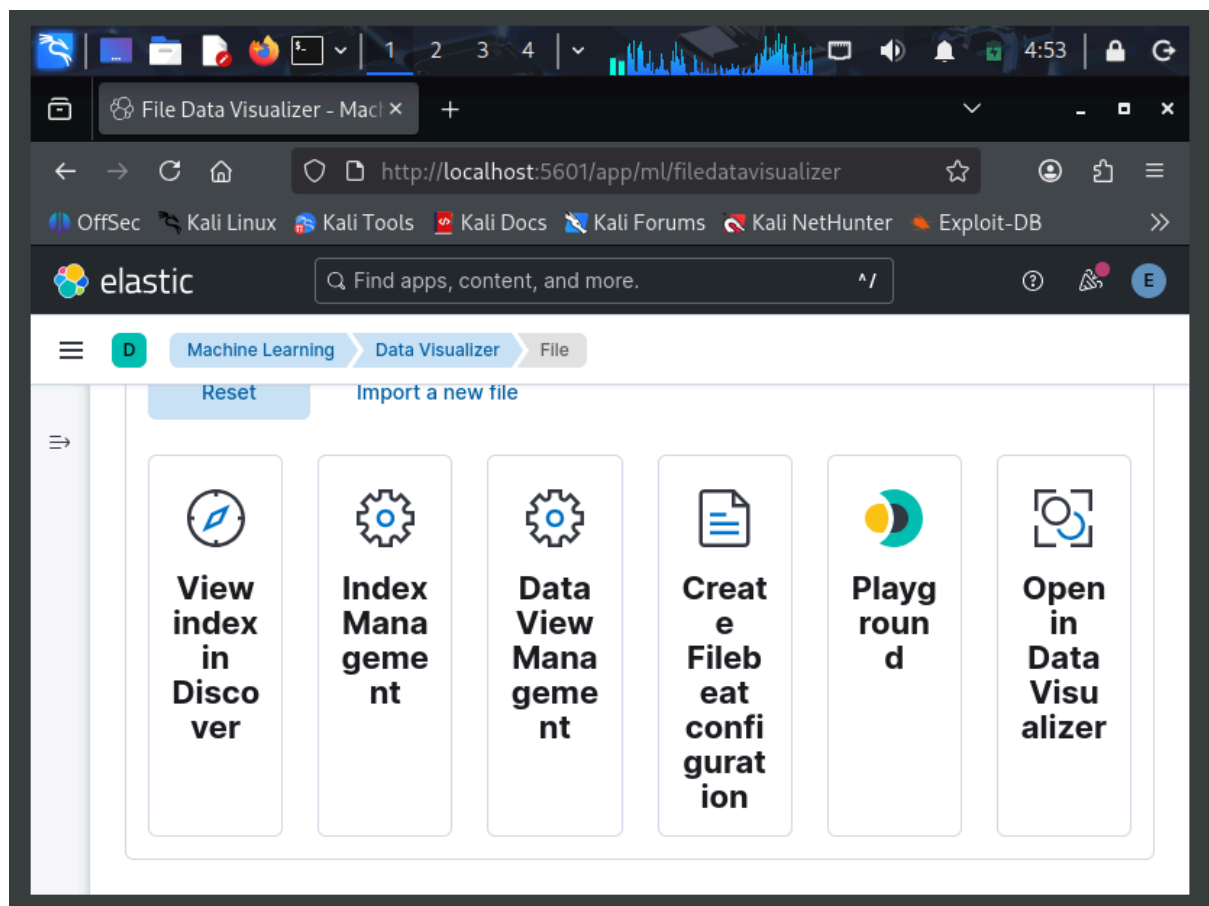


Figure 17: Kibana Discover (log correlation)

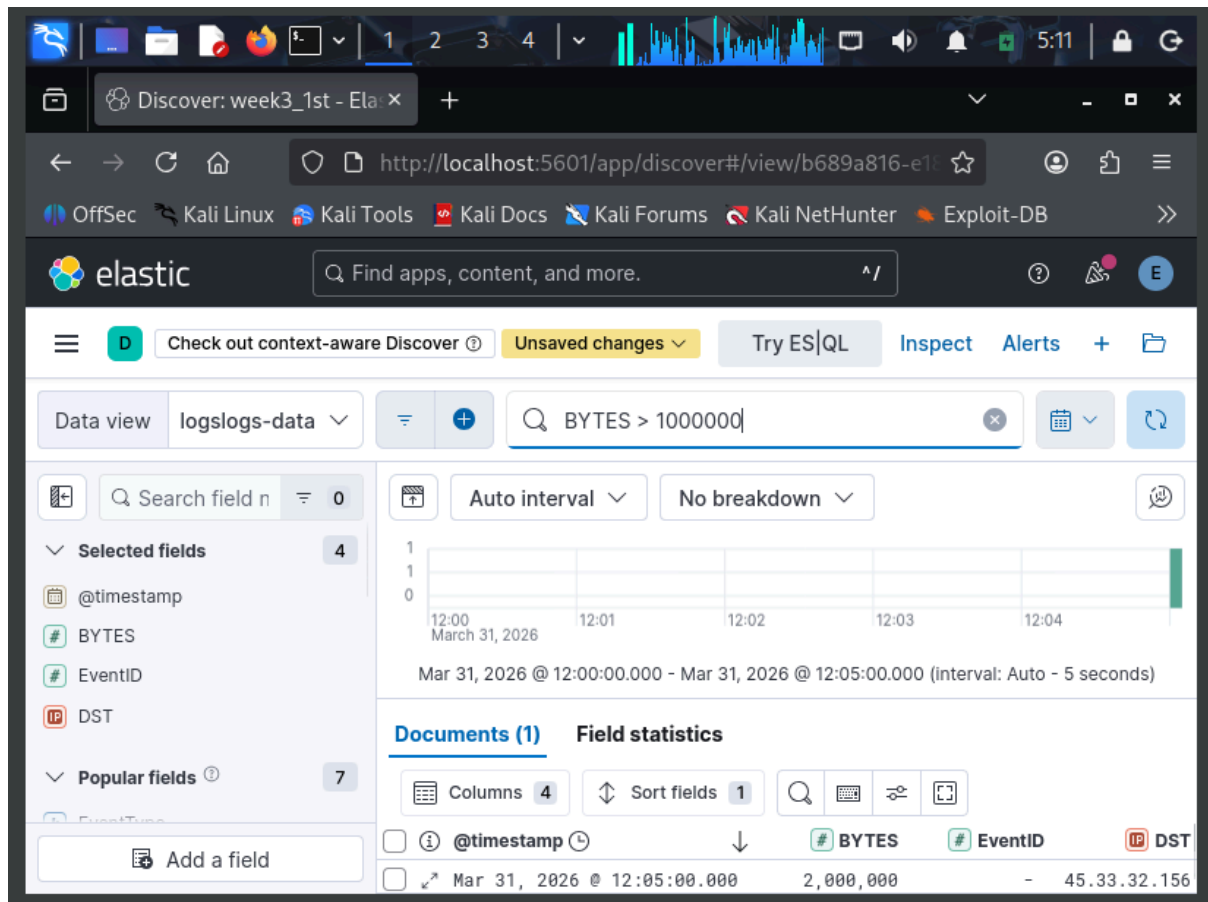


Figure 18: BYTES anomaly detection query

Due to tool limitations, response actions were simulated. The affected system was logically isolated, and the suspicious IP (45.33.32.156) was identified for blocking. Network monitoring was performed to prevent further data exfiltration.

The incident was escalated to Tier 2 using TheHive platform. A case was created documenting the attack details, affected systems, and observed indicators.

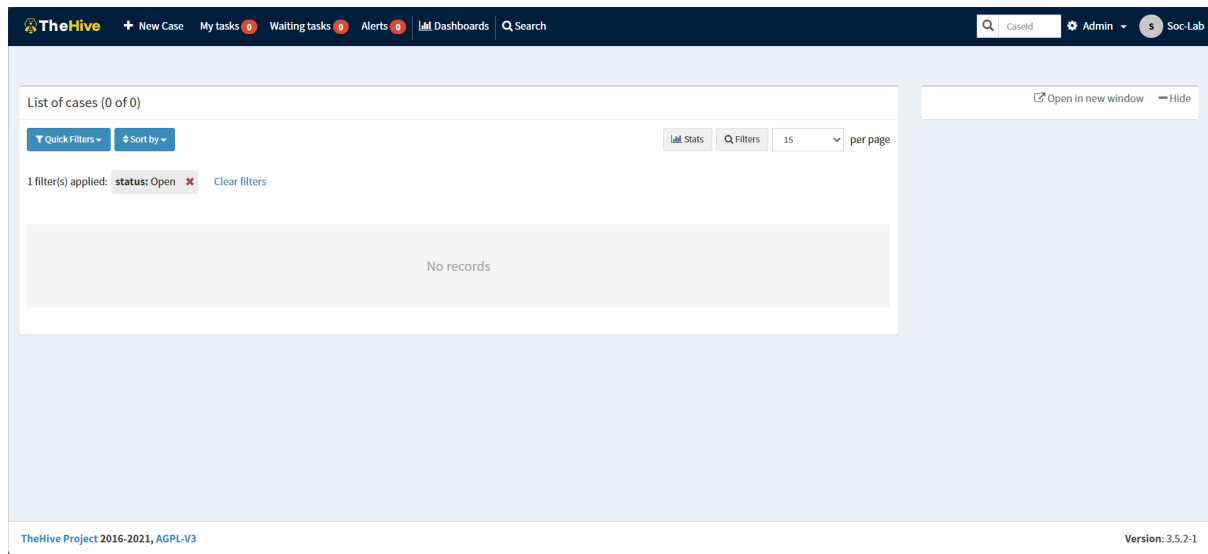


Figure 19: TheHive case creation screen

A high-priority incident was identified involving repeated failed login attempts followed by abnormal data transfer to an external IP. The activity indicates a potential brute-force attack followed by data exfiltration. The affected system was analyzed and monitored for further malicious behavior. The incident has been escalated to Tier 2 for deeper investigation. Immediate containment actions include isolating the affected host and monitoring outbound connections. The external IP has been flagged as suspicious. Further forensic analysis is recommended to determine the extent of compromise and prevent future occurrences.

The destination IP (45.33.32.156) was analyzed using VirusTotal and identified as suspicious, confirming potential malicious communication.

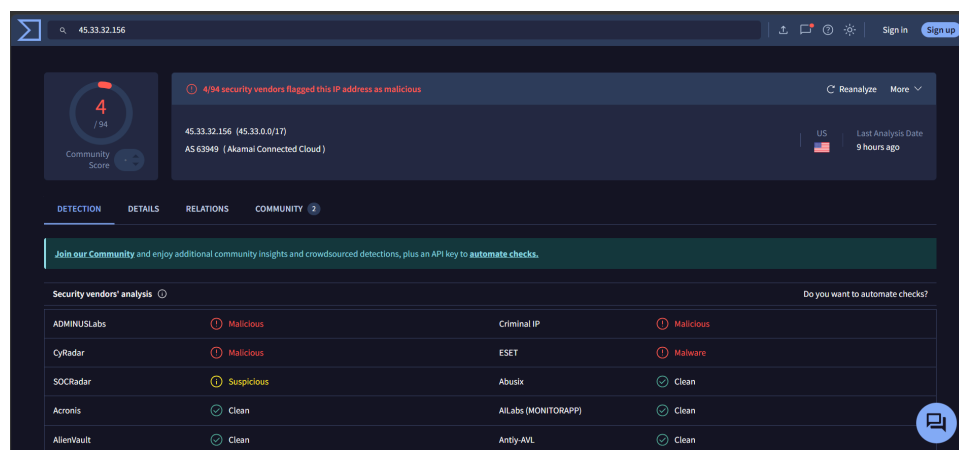
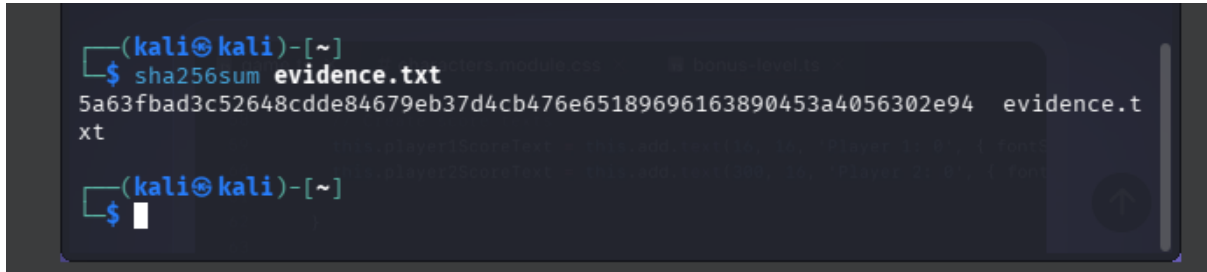


Figure 20: VirusTotal analysis



Network data was collected using netstat and stored securely. A SHA-256 hash was generated to ensure the integrity of the collected evidence.



```
(kali㉿kali)-[~]  
$ sha256sum evidence.txt  
5a63fbad3c52648cdde84679eb37d4cb476e65189696163890453a4056302e94  evidence.t  
xt  
(kali㉿kali)-[~]  
$
```

Figure 21: netstat output and hash